(and yes, they do cheat using fractional numbers for priority! But that's probably not the first time you see someone changing requirements after you pulled up a lot of work, is it?). After executing line #2 in listing 2.6, in fact, the list would be in this intermediate state (where heap properties are still violated.

Line #3 is then executed to reinstate heap property number 3, as we have seen in section 2.6.2, and the final result is shown in figure 2.9.B; notice the order in which the children of the root appear in the array: siblings are not ordered, and in fact a heap doesn't keep a total ordering on its elements, like a BST would.

2.6.4 Top

Now that we have seen how to insert a new element, let's define the `top` method, that will extract the heap's root and return it to the caller.

Figure 2.10 shows an example on a max-heap, highlighting the steps that are performed in listing 2.7.

The first thing we do is checking the heap is not empty: if it is, we can't certainly extract the top element, so we need to issue an error.

The idea is that, since we will remove the root of the heap, and by doing so we leave a "hole" in the array, we need to replace it with another element.

Listing 2.7 The top method